HMI ALARM / HISTORY / DATA LOGGING MAP

Data Group	PLC Source	HMI Screen	Logging Method	Trigger / Period	Retention
Current PLC State	M0100~M0114	Main / Auto Screen	Event Log	State Change	Project Log File
Safety Status	M0001, P00, P07	Main / Alarm Screen	Event Log	State Change	Project Log File
Active Alarm	D0310, M0120	Alarm / History Screen	Alarm Event Log	Alarm Set / Reset	Project Log File
Alarm History	ALM-001, ALM-101~108	Alarm / History Screen	CSV / TDMS Event Log	Alarm Occurrence	Project Log File
Test Result	D0020, D0021	Auto / Trend Screen	Cycle Result Log	Every Completed Cycle	Project Log File
Production Count	D0300~D0307	Main / Trend Screen	Periodic + Cycle Log	1 s / Cycle Complete	Project Log File
PLC I/O Status	P00~P0B, P20~P29	Main / Diagnostics Screen	Diagnostic Log	1 s or State Change	Project Log File
MCU Communication	D0023, ALM-108	Settings / Alarm Screen	Communication Log	1 s / Fault Event	Project Log File
HMI Setpoint Change	D0050~D0054	Settings Screen	Setpoint Audit Log	On Value Change	Project Log File
HMI Command Event	M0200~M0204	Alarm / History Screen	Operator Event Log	On Command Request	Project Log File

DATA LOG FILE FORMAT

CSV:
Timestamp, PLC_State, Test_Result,
Cycle_Count, PASS_Count, FAIL_Count,
RETEST_Count, Active_Alarm, MCU_Heartbeat

TDMS:
Process Trend, I/O Diagnostic,
Setpoint Change, Alarm Event History

ALM-001
SAFETY NOT READY
ALM-101
TRANSFER TO TEST TIMEOUT
ALM-102
CLAMP CLOSE TIMEOUT
ALM-103
TEST COMPLETE TIMEOUT
ALM-104
INVALID TEST RESULT
ALM-105
CLAMP RELEASE TIMEOUT
ALM-106
GATE POSITION TIMEOUT
ALM-107
OUTFEED TIMEOUT
ALM-108
MCU / MODBUS COMMUNICATION FAULT

[GENERAL HMI NOTES]

[1] HMI-01 SHALL PROVIDE SUPERVISORY MONITORING, SETPOINT ENTRY, ALARM HISTORY, AND DATA LOGGING.

[2] PLC-01 SHALL REMAIN THE PRIMARY CONTROL AUTHORITY. HMI-01 SHALL NOT BYPASS PLC OUTPUT INTERLOCKS.

[3] HMI START, STOP, RESET, MODE, AND COUNT RESET COMMANDS SHALL BE PROCESSED AS PLC REQUEST BITS ONLY.

[4] HMI COMMANDS SHALL NOT REPLACE HARDWIRED SAFETY FUNCTIONS INCLUDING ES-01, DR-01, SR-01, K1-A, OR K1-M.

[5] HMI-01 SHALL DISPLAY PLC STATE, INPUT/OUTPUT STATUS, ACTIVE ALARM, PRODUCTION COUNT, AND MCU COMMUNICATION STATUS.

[6] D0050~D0054 SETPOINT VALUES SHALL BE VALIDATED BY PLC-01 BEFORE USE IN AUTOMATIC OR MANUAL CONTROL LOGIC.

[7] PLC-01 SHALL UPDATE D0300~D0307 PRODUCTION COUNTERS AT EACH COMPLETED CYCLE.

[8] HMI-01 SHALL LOG ALARM EVENT, PROCESS RESULT, PRODUCTION COUNT, SETPOINT CHANGE, AND COMMUNICATION STATUS.

[9] HMI COMMAND REQUESTS M0200~M0204 SHALL BE RESET BY PLC-01 AFTER ACCEPTANCE, REJECTION, OR COMPLETION OF THE REQUEST.

[10] HMI USER LOGIN SHALL BE REQUIRED FOR PRODUCTION COUNT RESET AND PARAMETER CHANGE.

[11] PLC SEQUENCE, ALARM, TIMER, AND OUTPUT INTERLOCK:
SEE DWG 011 / DWG 012 / DWG 013.
PLC-01 / MCU-01 MODBUS RTU CIRCUIT: SEE DWG 014.

HMI SCREEN NAVIGATION, HMI/PLC COMMUNICATION FLOW, AND DATA OWNERSHIP ARCHITECTURE:
SEE DWG 015 / SHEET 1 OF 3.

NO.	DATE	REVISION
A	2026-09-16	INITIAL RELEASE

DWG TITLE
EDS Wafer Map Linked Package
Final Test and Multi-Bin Sorting Cell
SCADA HMI TAG AND COMMAND MAP

DRAWN BY: J. H. KO
DATE: 2026-09-16
DWG NO: 015
SHEET NO: 3 OF 3